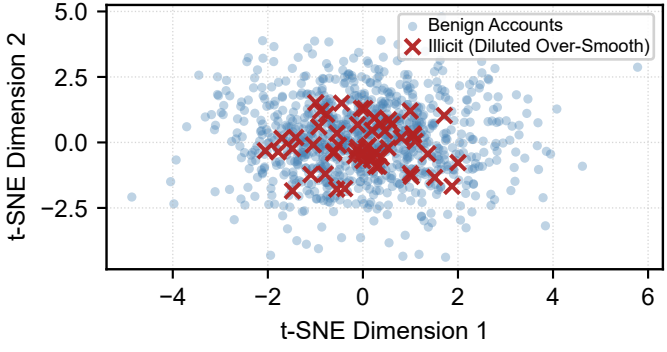
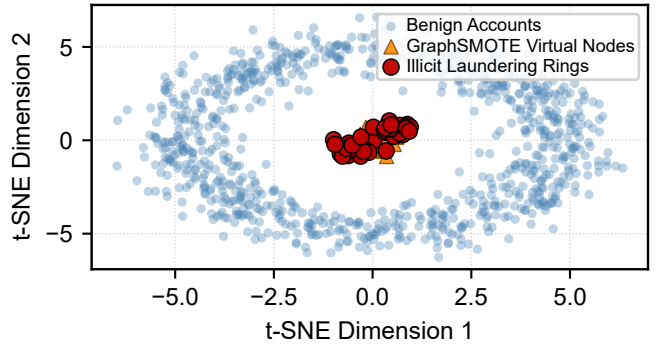

(a) Baseline GNN Layer 2 (Severe Over-Smoothing)
Silhouette S = -0.03, DB = 9.27
Benign Accounts
Illicit (Diluted Over-Smooth)
t-SNE Dimension 1
t-SNE Dimension 2
5.0
2.5
0.0
−2.5
−4
−2
0
2
4
6


(b) C-STGB Layer 2 (Typology Latent Manifolds)
Silhouette S = +0.78 ± 0.03, DB = 0.64
Benign Accounts
GraphSMOTE Virtual Nodes
Illicit Laundering Rings
t-SNE Dimension 1
t-SNE Dimension 2
5
0
−5
−5.0
−2.5
0.0
2.5
5.0